Terraform-Based Portfolio Website with Azure Blob Image Hosting

Project Overview:

This project demonstrates an end-to-end Azure Infrastructure as Code (IaC) implementation using Terraform. A single Azure Virtual Machine is provisioned to host a NGINX-based portfolio website. Additionally, an Azure Storage Account and Blob Container are created using Terraform backend configuration. A local image file is uploaded to the Blob Container entirely through Terraform code and consumed directly in the portfolio HTML page running on the VM. No manual uploads or portal-based configurations are performed.

Aim:

The objective of this project is to automate Azure infrastructure provisioning using Terraform, store static assets in Azure Blob Storage, and integrate those assets into a VM-hosted portfolio website without any manual intervention.

Architecture:

- Azure Resource Group
- Virtual Network and Subnet
- Azure Virtual Machine (Ubuntu Linux)
- Network Security Group allowing HTTP (Port 80)
- Azure Storage Account
  Azure Blob Container for static assets

Tools & Technologies Used:

- Microsoft Azure
- Terraform
- Ubuntu Linux
- NGINX Web Server
- Azure Blob Storage

HTML / CSS

Terraform Backend & Storage Configuration:

Terraform backend configuration is used to manage state securely. Using the same Terraform codebase, an Azure Storage Account and Blob Container are provisioned automatically. A local image file is uploaded to the Blob Container using Terraform resources, ensuring full automation without manual uploads via Azure Portal.

Kadali Gowtham Sai gowthamkadali2510@gmail.com

Terraform-Based Portfolio Website with Azure Blob Image Hosting

Virtual Machine & NGINX Setup:

After provisioning the VM with Terraform, NGINX is installed using Linux package manager commands.
The default web root directory
/var/www/html is overridden with custom portfolio HTML code.

Image Integration into Portfolio:

The image uploaded to Azure Blob Storage via Terraform is referenced directly in the portfolio HTML code by calling the Blob. The VM-hosted website dynamically loads the image from Azure Storage, demonstrating separation of compute and static asset storage.

Execution Steps:

• terraform init

```
PS C:\Users\gowth\OneDrive\Desktop\repo\Terraform-Based-Portfolio-Website-with-Azure-Blob-Image-Hosting> terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/azurerm...
```

• terraform validate

```
PS C:\Users\gowth\OneDrive\Desktop\repo\Terraform-Based-Portfolio-Website-with-Azure-Blob-Image-Hosting> terraform validate
```

• terraform plan

```
PS C:\Users\gowth\OneDrive\Desktop\repo\Terraform-Based-Portfolio-Website-with-Azure-Blob-Image-Hosting> terraform plan
azurerm_resource_group.rg: Refreshing state... [id=/subscriptions/b92e8a3c-9e1b-4643-9df9-bea61f286443/resourceGroups/go
wtham-rg]
azurerm_public_ip.pip: Refreshing state... [id=/subscriptions/b92e8a3c-9e1b-4643-9df9-bea61f286443/resourceGroups/gowtha
m-rg/providers/Microsoft.Network/publicIPAddresses/vm-pip]
azurerm_virtual_network.vnet: Refreshing state... [id=/subscriptions/b92e8a3c-9e1b-4643-9df9-bea61f286443/resourceGroups
/gowtham-rg/providers/Microsoft.Network/virtualNetworks/vnet1]
azurerm_network_security_group.nsg: Refreshing state... [id=/subscriptions/b92e8a3c-9e1b-4643-9df9-bea61f286443/resource
Groups/gowtham-rg/providers/Microsoft.Network/networkSecurityGroups/vm_nsg]
azurerm_storage_account.portfolio_sa: Refreshing state... [id=/subscriptions/b92e8a3c-9e1b-4643-9df9-bea61f286443/resour
ceGroups/gowtham-rg/providers/Microsoft.Storage/storageAccounts/gowthamstorageacc2457]
```

• terraform apply

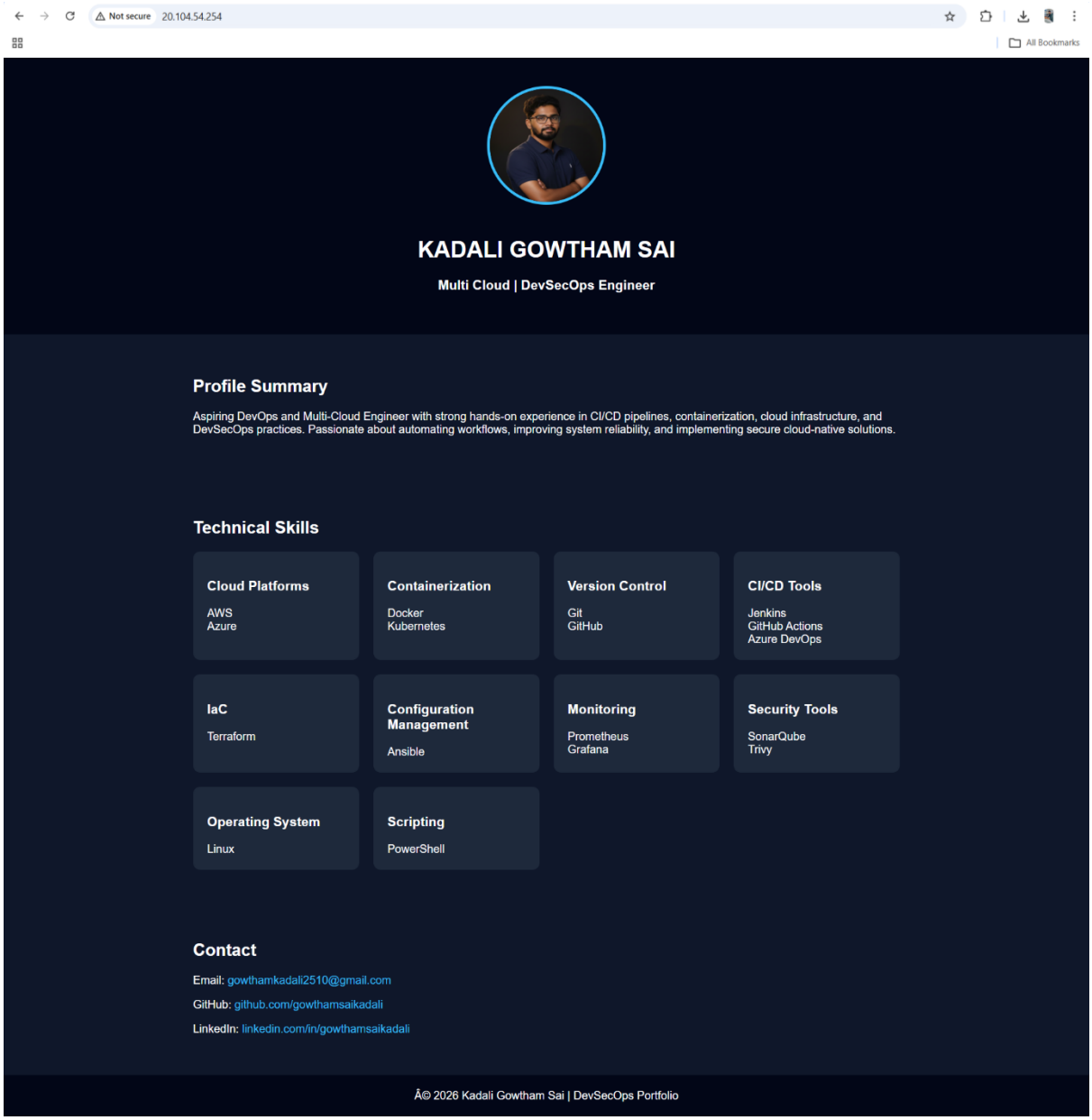
```
PS C:\Users\gowth\OneDrive\Desktop\repo\Terraform-Based-Portfolio-Website-with-Azure-Blob-Image-Hosting> terraform apply

azurerm_resource_group.rg: Refreshing state... [id=/subscriptions/b92e8a3c-9e1b-4643-9df9-bea61f286443/resourceGroups/go
wtham-rg]
azurerm_network_security_group.nsg: Refreshing state... [id=/subscriptions/b92e8a3c-9e1b-4643-9df9-bea61f286443/resource
Groups/gowtham-rg/providers/Microsoft.Network/networkSecurityGroups/vm_nsg]
azurerm_public_ip.pip: Refreshing state... [id=/subscriptions/b92e8a3c-9e1b-4643-9df9-bea61f286443/resourceGroups/gowtha
m-rg/providers/Microsoft.Network/publicIPAddresses/vm-pip]
azurerm_virtual_network.vnet: Refreshing state... [id=/subscriptions/b92e8a3c-9e1b-4643-9df9-bea61f286443/resourceGroups
/gowtham-rg/providers/Microsoft.Network/virtualNetworks/vnet1]
azurerm_storage_account.portfolio_sa: Refreshing state... [id=/subscriptions/b92e8a3c-9e1b-4643-9df9-bea61f286443/resour
ceGroups/gowtham-rg/providers/Microsoft.Storage/storageAccounts/gowthamstorageacc2457]
azurerm_subnet.subnet: Refreshing state... [id=/subscriptions/b92e8a3c-9e1b-4643-9df9-bea61f286443/resourceGroups/gowtha
```

Output & Verification:

The portfolio website is accessed using the public IP address of the Azure VM. The image stored in Azure Blob Storage is successfully rendered on the website, confirming correct Terraform-based provisioning and integration.

Terraform-Based Portfolio Website with Azure Blob Image Hosting



Conclusion:

This project demonstrates a real-world DevOps workflow using Terraform to manage both compute and storage resources in Azure. It highlights automation, backend state management, and best practices by eliminating manual configuration and uploads entirely.